This Patient Group Direction (PGD)is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Patient Group Direction
for the administration of Acetazolamide for the prevention of Altitude Sickness

Patient Group Direction, version 006, issued 30 September 2026. Supersedes Version 005, 14 September 2026.

Owner and authorising organisation: Get Real Health Limited, company number 12744898, registered office Zenvite Health Unit 23, St. Asaph Business Park, St. Asaph, Denbighshire, LL17 0LJ. This Patient Group Direction is authorised on behalf of Get Real Health Limited by its Medical Director and its Head Pharmacist, whose signatures for this version are on the authorisation page at the end of the document. It applies to the registered healthcare professionals named under it by a pharmacy or clinic that has adopted it through a Get Real Health service agreement, at the premises stated in that adoption, and to no other organisation.

Summary of the governing guidance for altitude sickness: BNF acetazolamide monograph (unlicensed indication), NaTHNaC / TravelHealthPro altitude illness factsheet, and the Wilderness Medical Society clinical practice guidelines for acute altitude illness, 2024 update. There is no NICE or NICE CKS topic for altitude sickness.

Overview

- **Altitude sickness** (acute mountain sickness or AMS) occurs when people ascend to **high altitudes too quickly**, typically **above 2,500 metres**.
- Caused by **hypoxia** due to reduced atmospheric pressure.

GET REAL HEALTH

Types of Altitude Illness

1. **Acute Mountain Sickness (AMS)** – common and usually mild
2. **High-Altitude Cerebral Oedema (HACE)** – life-threatening
3. **High-Altitude Pulmonary Oedema (HAPE)** – life-threatening

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Symptoms of AMS

- Develops within **6–12 hours** of ascent
- Headache (required for diagnosis) **plus** one or more:
  - Nausea or vomiting
  - Dizziness or light-headedness
  - Fatigue or weakness
  - Poor sleep
  - Loss of appetite

Prevention Advice

1. Gradual Ascent

- Ascend **no more than 300–500 metres per day** above 2,500 m
- Include a **rest day every 3–4 days** or for every 1,000 m ascent

2. Acetazolamide (for prevention)

- Consider for:
  - Previous AMS
  - Rapid or unavoidable ascent
  - High-risk individuals

Dose:

- **Acetazolamide 125 mg twice daily, starting 1 to 2 days before ascent and continuing for 2 days at the highest altitude, or until descent begins (Wilderness Medical Society 2024: 125 mg twice daily is as effective as 250 mg for prevention with fewer adverse effects)**

Acetazolamide is **not licensed in the UK** for AMS, but widely used **off-label**.

Management of Mild AMS

- **Do not ascend further** until symptoms resolve
- Rest and monitor
- Treat headache with **paracetamol or ibuprofen**

- Mild symptoms usually resolve within **24–48 hours**

Acetazolamide for treatment

- **250 mg twice daily until symptoms resolve, together with rest or descent; descent is the definitive treatment and acetazolamide is an adjunct**

Management of Severe AMS / Suspected HACE or HAPE

- **Immediate descent** is the **priority**
- Administer **supplemental oxygen** if available
- Consider **dexamethasone** (HACE):
  - **8 mg initially**, then **4 mg every 6 hours**
- Consider **nifedipine** (HAPE):
  - **20 mg slow release BD**, only under specialist advice

These are emergency situations: urgent medical attention and descent are critical. Dexamethasone and nifedipine are not supplied under this PGD; they are listed so that the practitioner recognises the emergency and directs the patient to urgent medical care.

When to Seek Medical Attention

- Symptoms worsen or don't improve within 24 hours
- Neurological symptoms (e.g. confusion, ataxia)
- Breathing difficulty at rest (possible HAPE)
- Severe fatigue, reduced consciousness

Other Advice

- Stay well hydrated
- Avoid alcohol
- Avoid sleeping pills or sedatives
- Educate travellers on recognising and responding to symptoms early

Patient Group Direction

for the supply/administration of

Acetazolamide

for the treatment/prevention of

Altitude Sickness

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	• Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines ● All users must have completed training in working under Patient Group Directions, for example the CPPE PGD e-learning and e-assessment or an equivalent, and have been assessed as competent to work under a PGD before first working under this one; previous experience of working under a PGD is not required ● All users should complete the Get Real Health training for this PGD before first working under it, or have completed equivalent training elsewhere, for example another provider's training for the same service or substantial recent experience of delivering it under a PGD, and must read this PGD in full before first working under it. The date the training was completed, and for equivalent training what it was, is to be recorded in the Get Real Health training record or in the employing organisation's own training record. The employing organisation remains responsible for being satisfied that each practitioner it authorises under this PGD is competent to do so. ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service

Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Acetazolamide is indicated for the prevention and treatment of acute mountain sickness (AMS) in adults travelling to altitudes above 2,500 metres. For prevention, treatment is started before ascent. For treatment, it is started at symptom onset. Use for AMS is OUTSIDE the marketing authorisation (the SmPC indications are glaucoma, fluid retention and epilepsy); it is supported by the BNF and Wilderness Medical Society guidance and is supplied off-label under this PGD. Tell the patient this as part of consent and record that you did.
Inclusion criteria	- Adults aged 18 years and over. - Travelling to or currently at altitudes above 2,500 metres. - Requesting preventative treatment or symptomatic treatment for AMS. - Informed consent obtained. - No contraindications to acetazolamide therapy.
Exclusion criteria	- Known hypersensitivity to acetazolamide or sulfonamides. - Severe renal or hepatic impairment, or hepatic cirrhosis. - Adrenal insufficiency (Addison's disease). - Hyperchloraemic (metabolic) acidosis, hypokalaemia or hyponatraemia, or a history of electrolyte imbalance. - Previous non-cardiogenic pulmonary oedema after acetazolamide. - Pregnancy or breastfeeding. - Taking lithium, phenytoin, high-dose aspirin or a potassium-depleting diuretic. Refer.
Cautions	- Use with caution in patients with mild renal impairment. - Monitor for symptoms of electrolyte imbalance and dehydration. - Advise patient to maintain adequate hydration and avoid alcohol. - Discontinue if severe side effects develop.
Actions if patient is excluded or declines treatment	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP as appropriate

Details of the medicine

Name, form and strength of medicine	Acetazolamide 250 mg tablets (scored)
Legal category	POM
Storage	Store below 25°C in a dry place.
Route/method of administration	Oral administration, with or without food.
Dose and frequency	Prevention: 125 mg (half a 250 mg tablet) twice daily, starting 1 to 2 days before ascent and continuing for 2 days after reaching the highest altitude, or until descent begins. Treatment: 250 mg twice daily for up to 3 days. Acetazolamide is not a substitute for descent.

Quantity to be administered and/or supplied	Prevention: half a tablet (125 mg) twice daily for (1 to 2 lead-in days + days ascending + 2 days), rounded up to whole tablets, MAXIMUM 14 tablets (28 doses, 14 days). Treatment: 6 tablets (250 mg twice daily for 3 days), which may be supplied in addition to the prevention course where the itinerary makes descent difficult; maximum total 20 tablets per supply.
Maximum or minimum treatment period	Prevention: maximum 14 days per supply without review. Treatment: maximum 3 days (6 tablets). A treatment course supplied alongside a prevention course is a standby course: it is taken only if symptoms develop, for no more than 3 days, and no further treatment supply is made under this PGD without review.
Adverse effects	Paresthesia, polyuria, taste alteration, nausea, drowsiness. Rare: electrolyte imbalance, kidney stones, blood dyscrasias. Refer to SmPC for full details.

Yellow card reporting	Report suspected adverse effects via https://yellowcard.mhra.gov.uk and advise the traveller to consult a healthcare professional if symptoms worsen.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication ● date of supply, dose, form and route, quantity supplied, batch number and expiry date ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and the actions taken, and that the supply was made under this PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medication.

Follow-up advice to be given to patient or carer	Do not ascend further while symptomatic. Descend and seek help urgently for symptoms not improving within 24 hours, or for confusion, unsteadiness, severe headache, breathlessness at rest, cough with frothy sputum or reduced consciousness (possible HACE or HAPE). Acetazolamide is not a substitute for descent.

Key references

- NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions. Published March 2017 https://www.nice.org.uk/guidance/mg2
- NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 https://www.nice.org.uk/guidance/mpg2/resources
- Wilderness Medical Society Clinical Practice Guidelines for the Prevention, Diagnosis and Treatment of Acute Altitude Illness: 2024 Update (Wilderness & Environmental Medicine, 2024); NaTHNaC TravelHealthPro factsheet, Altitude illness; BNF, acetazolamide
- British National Formulary (BNF) Key information on the selection, prescribing, dispensing and administration of medicines BNF (British National Formulary) | NICE
- Summary of Product Characteristics

Agreement to practise by the Registered Healthcare Professional

By signing this PGD you are confirming that you agree to its contents and that you will work within it. PGDs do not remove inherent professional obligations or accountability. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of pharmacy premises / clinic premises to which this PGD relates			
Name of healthcare professional	Job title	Registration number	Signature
Valid from date		Expiry date	

30/9/26	31/7/27

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or vaccine.

This section must be signed by the superintendent or clinical lead responsible for this service. To be completed by the adopting pharmacy. These are the adopting organisation's own signatories and must not be Get Real Health staff.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date
Name of professional group signatory	Job title & name of organisation	Signature	Date

Change history to version 001

Later versions are recorded in the Version and change record at the end of this document.

Version number	Change details	Date
001	Development & issue of new PGD	25th June 2025

Version and change record

Version: v006

Issued: 30 September 2026

Supersedes: Version 005, 14 September 2026

Valid from: 30 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Training before first use, applied to every Get Real Health PGD on 30 September 2026: the staff section now states that each practitioner should complete the Get Real Health training for this PGD, or equivalent training elsewhere, before first working under it, must read the PGD in full before first use, and that the date of completion is recorded. Training is recorded; it is not a condition of access, and the employing organisation remains responsible for being satisfied of competence. No clinical content changes.

Previous version records

Version: v005

Issued: 14 September 2026

Supersedes: Version 004, 11 September 2026

Valid from: 14 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Identity and training, applied to every Get Real Health PGD on 14 September 2026 after two adopting pharmacies wrote: the Get Real Health logo is in the page header; the organisation that owns and authorises this direction, Get Real Health Limited, is named under the strapline with its company number and registered office; the requirement that practitioners must previously have used a PGD is replaced by a requirement for PGD training and a competence assessment, so that a newly registered pharmacist or technician who has completed the training may work under it; and the earlier change history table is headed with the versions it covers.

Previous version records

Version: v004

Issued: 11 September 2026

Supersedes: Version 003, 11 September 2026

Valid from: 11 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

The guidance summary now states that dexamethasone and nifedipine are not supplied under this PGD and are listed only so the emergency is recognised.

The maximum treatment period row now says that a treatment course supplied with prevention is a standby course, taken only if symptoms develop and for no more than 3 days.

The records row has its punctuation restored and now asks for the batch number and expiry date.

Wording corrections from the tool alignment and adversarial review of the consultation tools, 11 September 2026. Each correction resolves a contradiction inside the document or a plain error, taking the safer reading; no indication is widened, no dose raised and no exclusion removed. Questions that need a clinical decision are recorded separately and are not changed here.

Previous version records

Version: v003

Issued: 11 September 2026

Supersedes: Version 002, 10 September 2026

Valid from: 11 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

The guidance summary is re-attributed to its real sources (BNF, NaTHNaC, Wilderness Medical Society 2024). There is no NICE CKS topic on altitude sickness; the previous version said there was. The summary's prevention dose is corrected to 125 mg twice daily, consistent with the PGD's own dose row.

Maximum quantity set to 14 tablets for prevention (28 doses, 14 days, matching the maximum period) and 6 for treatment, total 20; the previous 28-tablet maximum was double the stated maximum period.

Structural clean-up applied to every live Get Real Health PGD on 11 September 2026, following the adversarial review of 10 September: the adopting pharmacy's authorisation block is blank, for the pharmacy's own signatories; one signed authorisation page for this version replaces the earlier signature blocks; every validity cell states this version's dates (valid from 11 September 2026, expiry 31 July 2027); narration about earlier versions, the consultation tool and individual customers is removed from the body of the document; the change records of earlier reissues are carried below as previous version records. No clinical criterion changes unless listed.

Previous version records

Version: v002

Issued: 10 September 2026

Supersedes: Version 001, 25 June 2025

Valid from: 10 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Validity: this version is valid from 10 September 2026 and expires on 31 July 2027. The signed master previously stated no period of validity, or an expiry of 31 October 2026; those are superseded.

Off-label use declared: acetazolamide is not licensed for acute mountain sickness. The indication now says so, names the supporting guidance, and requires the patient to be told.

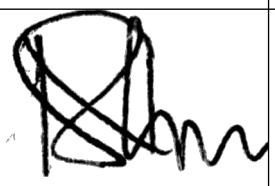
Exclusion criteria: SmPC contraindications added (adrenal insufficiency, hyperchloraemic acidosis, hepatic cirrhosis, previous pulmonary oedema on acetazolamide) and the undefined 'other interacting medications' replaced by the SmPC list (lithium, phenytoin, high-dose aspirin, potassium-depleting diuretics).

Details of the medicine: the form and strength (250 mg scored tablets), the quantity and the maximum treatment period were absent, which Schedule 16 requires. The prevention dose of 125 mg is now stated to be half a tablet.

Follow-up advice: the previous text was a skin-infection template ('do not improve in 3 to 4 weeks'). Replaced with the altitude advice from the document's own guidance summary: do not ascend while symptomatic, descend for symptoms not improving in 24 hours or for any sign of HACE or HAPE, acetazolamide is not a substitute for descent.

Signed on behalf of Get Real Health

Version v006, 30 September 2026.

Name	Qualification	Signature	Date
Nitin Shori	Medical Director GMC: 6047293		30 September 2026
Chris Pilkington	Head Pharmacist GPhC: 2046322		30 September 2026

Both authorising signatories reviewed this version together on 30 September 2026 and gave their authorisation for it to be issued. Signatures were applied digitally on their joint instruction, which is the established process for Get Real Health PGDs. This version is not valid without both signatures.